Vito Mengers

PhD Student in Robotics & Interdisciplinary Intelligence Research

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Research

My research centers on the Active Interconnect (AICON) framework in which internal models get continuously refined based on sensory input. Initially explored for robust robotic perception, I demonstrated that the framework can also drive action selection through gradient descent. This tight coupling of perception and action lets intelligent agents both react adaptively to environmental changes and actively probe their surroundings to improve perception.

Beyond robotics, I study the AICON framework as a model for biological behavior, showing that the same architecture can capture processes such as human visual attention, human sequential problem solving, and collective behavior. The resulting systems are rich, robust, and explainable, offering insight into the interactions between recursive modules—both as computational tools and as models of cognitive mechanisms.

Education

TU Berlin / Science of Intelligence

Berlin, Germany
2021–present

PhD – Robotics & Interdisciplinary Intelligence Research

Expected submission – March 2027.

TU Berlin

Berlin, Germany 2018–2021

M.Sc. – Computer Engineering

- Grade – 1.1 (ECTS Rank A)
- Honors – One of the best graduates of the year

TU Berlin

Berlin, Germany 2015–2019

B.Sc. – Computer Engineering

- Grade – 1.3 (ECTS Rank A)
- Honors – One of the best graduates of the year

Journal Articles

- **V. Mengers**[*], N. Roth[*], O. Brock, K. Obermayer, and M. Rolfs. *A robotics-inspired scanpath model reveals the importance of uncertainty and semantic object cues for gaze guidance in dynamic scenes.* Journal of Vision, 2025. [PDF, Project Page]
- **V. Mengers**[*], M. Raoufi[*], O. Brock, H. Hamann, and P. Romanczuk. *Leveraging uncertainty in collective opinion dynamics with heterogeneity.* Scientific Reports, 2024. [PDF, Project Page]

Conference Papers

- P. Lowin, **V. Mengers**, and O. Brock. *Coupled Particle Filters for Robust Affordance Estimation.* International Conference on Robotics and Automation (ICRA), 2026. [PDF, Project Page]
- **V. Mengers**, B. D. Cao, and O. Brock. *A Mechanistic Model for Collective Motion from Sensorimotor Regularities.* International Conference on the Simulation of Adaptive Behavior (SAB), 2026. [PDF, Project Page]
- M. Migacev[*], **V. Mengers**[*], A. Köngeter, and O. Brock. *No Plan, Yet Human: A Reactive Robotics Model Predicts Human Planning Failures on a Clinical Task.* International Conference on the Simulation of Adaptive Behavior (SAB), 2026. [PDF, Project Page]
- D. Burema[*] and **V. Mengers**[*]. *Whose "Intelligence" Are We Building? On Exclusion in Conceptual Frameworks of Artificial Intelligence.* AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026. [PDF]
- **V. Mengers** and O. Brock. *No Plan but Everything Under Control: Robustly Solving Sequential Tasks with Dynamically Composed Gradient Descent.* International Conference on Robotics and Automation (ICRA), 2025. [PDF, Project Page] **ICRA Best Paper in Planning and Control**

[*]Equal contribution.

- A. Pfisterer*, X. Li*, **V. Mengers**, and O. Brock. *A Helping (Human) Hand in Kinematic Structure Estimation.* International Conference on Robotics and Automation (ICRA), 2025. [PDF, Project Page]
- **V. Mengers**, A. Battaje, M. Baum, and O. Brock. *Combining Motion and Appearance for Robust Probabilistic Object Segmentation in Real Time.* International Conference on Robotics and Automation (ICRA), 2023. [PDF]

Workshop Papers & Extended Abstracts

- **V. Mengers** and O. Brock. *When Gradients Are Enough: Non-Stationary Potential Fields for Reactive Control.* German Robotics Conference (GRC), 2026. [PDF]
- **V. Mengers***, A. Koenig*, X. Li*, A. Sieler, A. Battaje, and O. Brock. *Stop Merging, Start Separating: Why Merging Learning and Modeling Won't Solve Manipulation but Separating the General From the Specific Will.* ICRA Workshop: Learning Meets Model-Based Methods for Contact-Rich Manipulation, 2025. [PDF]
- **V. Mengers***, A. Battaje*, and O. Brock. *AICON: A Representation for Adaptive Behavior.* German Robotics Conference (GRC), 2025. [PDF]
- A. Pfisterer*, X. Li*, **V. Mengers**, and O. Brock. *A Handy Solution to Kinematic Structure Estimation.* German Robotics Conference (GRC), 2025. [PDF]

Preprints & Under Review

- **V. Mengers** and O. Brock. *Resolving Conflicts Where and When They Arise: Reactive Composition of Multi-Goal Behavior.* arXiv Preprint, 2026. [PDF]
- A. Battaje, M. Bernhard, **V. Mengers**, and O. Brock. *Building Generalization Into Behavior Generation Via Adaptive Compositions of Regularities.* arXiv Preprint, 2026. [PDF]
- E. Sebastián, A. Pfisterer, **V. Mengers**, O. Brock, and A. Prorok. *World-Task Factorization for Robot Learning.* arXiv Preprint, 2026. [PDF]

Experience

Robotics and Biology Lab (Oliver Brock) – TU Berlin
Berlin, Germany

Research Assistant
2021–present

Developing computational models of intelligent behavior for robots. Furthermore, involved in teaching of multiple masters-level courses on robotics.

Research Cluster "Science of Intelligence"
Berlin, Germany

PhD Student Member
2021–present

Developing computational models of intelligent behavior in biology. Furthermore, involved in organization of multiple interdisciplinary projects, events, as well as being part in the ethics committee of the cluster.

Awards and Scholarships

IEEE ICRA Best Paper Award on Planning and Control
2025

Finalist for Best Paper (Overall), Best Student Paper, and Best Paper on Robot Learning.
Vito Mengers & Oliver Brock, No Plan but Everything under Control - Robustly Solving Sequential Tasks with Dynamically Composed Gradient Descent

"For challenging the planning paradigm by introducing a feedback-driven method that solves sequential tasks through continuous adaptation."

Studienstiftung des deutschen Volkes Scholarship (4 years)
2018

Studienstiftung des deutschen Volkes (German Academic Scholarship Foundation)

Prestigious German national scholarship awarded to highly talented students demonstrating exceptional academic achievement and societal commitment.

Deutschlandstipendium Scholarship (2 years)
2016

Deutschlandstipendium (awarded by TU Berlin, financed by the German Federal Government, Siemens, and CARIAD)

Merit-based scholarship awarded for excellent academic performance and potential, funded jointly by the German Federal Government and private industry partners.

Academic Service

Submissions Chair & Publication Chair Berlin, Germany 2026
18th International Conference on the Simulation of Adaptive Behavior (SAB 2026)

Workshop Co-Organizer Atlanta, GA, USA 2025
1st Workshop on Acoustic Sensing and Representations for Robotics (RoboAcoustics) at ICRA25

Summerschool Co-Organizer - Responsible for Poster Sessions Berlin, Germany 2023
SCIoI & ISAB Summerschool

Symposium Co-Organizer Berlin, Germany 2022
3rd PhD-PostDoc Symposium of "Science of Intelligence"

Regular Reviewer
IEEE International Conference on Robotics and Automation (ICRA), Robotics Science and Systems (RSS), International Conference on Simulation of Adaptive Behavior (SAB), IEEE Robotics and Automation Letters (RA-L), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES), Conference on Robot Learning (CoRL), IEEE Transactions on Robot Learning (T-RL)

Student Supervision

Bao Duc Cao (Master's Thesis) 2026
Supervised and Examined by Oliver Brock & Pawel Romanczuk

Antonia Köngeter (Master's Thesis) 2025
Supervised and Examined by Oliver Brock & Eva Wiese

Malte Bernhard (Master's Thesis) 2025
Co-Advised with Aravind Battaje & Alexander Koenig; Supervised and Examined by Oliver Brock & Marc Toussaint

Adrian Pfisterer (Master's Thesis) 2024
Co-Advised with Xing Li; Supervised and Examined by Oliver Brock & Guillermo Gallego; ***Winner of the Rolf-Niedermeier-Award***

Patrick Lowin (Master's Thesis) 2024
Supervised and Examined by Oliver Brock & Guillermo Gallego

Teaching

Advisor and Organizer — Course "Robotics Seminar" 2026
Administrative Organization, Overarching Topic Conception, and Supervising student presentations and writing, facilitated seminar discussions.

Responsible for Exercises & Lecture — Motion Planning 2025
Lecture delivery and exercise supervision in course "Robotics".

Advisor — Course "Robotics Project" 2025
Supervised student projects, guided presentations and report writing, provided technical mentorship.

Responsible for Exercises & Lecture — Advanced Robot Control 2024
Lecture delivery and exercise supervision in course "Robotics".

Advisor — Course "Robotics Seminar" 2024
Supervised student presentations and writing, facilitated seminar discussions.

Responsible for Exercises — Visual Servoing 2023
Exercise instructor and assessor for course "Robotics".

Responsible for Exercises — Recursive Estimation 2022
Exercise instructor and assessor for course "Robotics".

Advisor — Course "Robotics Project" 2022
Supervised student projects, guided presentations and report writing, provided technical mentorship.

Advisor — Course "Robotics Seminar" 2022

Supervised student presentations and writing, facilitated seminar discussions.

Responsible for Exercises — Basics of Robot Control 2021

Exercise instructor and assessor for course "Robotics".

Talks & Posters

- **Poster** @ ICRA26 — *Coupled Particle Filters for Robust Affordance Estimation* (Together with Patrick Lowin) (2026)
- **Poster** @ GRC26 — *When Gradients Are Enough: Non-Stationary Potential Fields for Reactive Control* (2026)
- **Poster** @ ICRA25 Workshop "Beyond the Lab: Robust Planning and Control in Real World Scenarios" — *No Plan but Everything Under Control* (2025)
- **Talk & Poster** @ ICRA25 — *No Plan but Everything Under Control: Robustly Solving Sequential Tasks with Dynamically Composed Gradient Descent* (2025)
- **Poster** @ ICRA25 Workshop "Learning Meets Model-Based Methods for Contact-Rich Manipulation" — *Stop Merging, Start Separating: Why Merging Learning and Modeling Won't Solve Manipulation but Separating the General From the Specific Will* (2025)
- **Poster** @ GRC25 — *AICON: A Representation for Adaptive Behavior* (2025)
- **Poster** @ GRC25 — *A Handy Solution to Kinematic Structure Estimation* – Together with Adrian Pfisterer (2025)
- **Talk & Poster** @ SCIoI's Scientific Networking Days — *Gradient Descent through Interconnected Recursive Estimators* (2024)
- **Talk** @ NetSci24 — *Centrality-Certainty Correlation in Heterogeneous Opinion Dynamics* – Together with Mohsen Raoufi (2024)
- **Talk** @ SCIoI's Thursday Morning Talks — *Principles at Play: What is Intelligence?* – Together with Aravind Battaje (2023)
- **Talk** @ SCIoI's Scientific Networking Days — *Interdisciplinary Applications of Interconnected Recursive Estimators* (2023)
- **Invited Talk** @ IRIDIA Seminar (Université Libre de Bruxelles) — *From Segmentation to Collaboration: Exploring the Role of Robust Estimation of Object Segmentation in Achieving Robust Collective Estimation* (2023)
- **Poster** @ ICRA23 — *Combining Motion and Appearance for Robust Probabilistic Object Segmentation in Real Time* – Together with Manuel Baum (2023)
- **Talk** @ SCIoI's Scientific Networking Days — *Differentiable Interconnected Recursive Estimation* (2022)

Outreach & Public Engagement

Podcast Interview About My Research (in German) 2025

Podcast of the DFG (German Research Foundation) "Exzellent Erklärt"

Science Slam at the Berlin Science Week 2025

Title of my Slam – "Why my robot is still stupid"

Interactive Presentation and Demo at the Alexander von Humboldt Foundation's Annual Alumni Event 2024

Interdisciplinary Exchanges & Public Outreach

Girl's Day at Science of Intelligence 2023

Full-day event to inspire German school girls to pursue a career in science

Participation in "I'm a Scientist – Get Me Out of Here!" 2022

Participation in a nationwide dialogue initiative connecting school students directly with researchers